

Gookho Song

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Research Interests

Core fields: metasurfaces · wavefront shaping · computational optics · physics-guided deep learning · holographic imaging · Fourier ptychography · computational spectroscopy & hyperspectral imaging.
I build optical encoders (disordered and inverse-designed metasurfaces, scattering media) paired with physics-informed reconstruction to enable compact, snapshot multimodal imaging (spectrum, phase and depth) that conventional optics cannot achieve.

Education

Korea Advanced Institute of Science and Technology (KAIST) <i>Ph.D. Candidate, Bio and Brain Engineering</i>	2022–present Daejeon, Korea
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- **Advisor:** Prof. Mooseok Jang
- **Focus:** metasurfaces; multimodal (hyperspectral, phase, depth) imaging; random/computational encoding

Korea Advanced Institute of Science and Technology (KAIST) <i>M.S., Bio and Brain Engineering · GPA 4.04 / 4.3</i>	2020–2022 Daejeon, Korea
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- **Advisor:** Prof. Mooseok Jang
- **Thesis:** *Differential detection of ultrasound-modulated light for deep tissue imaging*

Inha University <i>B.S., Electronic Engineering, double major in Biotechnology · GPA 4.11 / 4.5</i>	2016–2020 Incheon, Korea
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Honors & Awards

Student Paper Prize — Optica Imaging Congress, Computational Optical Sensing and Imaging (COSI) · Seattle, WA, USA	Aug 2025
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- Sole winner in the COSI track, awarded for an oral presentation.

Best Student Paper Award, 1st Place — Next-Generation Lithography & Patterning Conference · Korea	2024
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- 1st place; research grant of ₩1,000,000. Topic: metasurface wavefront sensor for semiconductor metrology.

Best Presentation Award — KAIST Optics Group Student Poster Competition	Jul 2024
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- Topic: compact spectrometer based on a double-layer disordered metasurface.

Journal Articles

G. Song = applicant · [†] equal contribution · ^{*} corresponding author

Google Scholar (Sep 2026): 231 citations · h-index 8 · i10-index 7

1. Y. Kim[†], **G. Song**[†], H. Kim, C. Lee, M. Jang^{*}, “Single-shot imaging of phase objects fully enclosed by dynamic scattering layers,” *Optica*, vol. 13, no. 7, 1245–1253, **2026**. doi:10.1364/OPTICA.593328
2. S. Park, D. Lee, N. Kim, W. Tian, **G. Song**, M. Jang, “Photonics Breakthroughs 2025: Reconstructive spectrometer using double-layer disordered metasurfaces,” *IEEE Photonics Journal*, vol. 18, no. 4, 5100307, **2026**. (*Photonics Breakthroughs 2025*) doi:10.1109/JPHOT.2026.3696089
3. T. Yin, **G. Song**, D. Lee, J. Oh, M. Jang, “Impact of system–model mismatch in Fourier ptychographic microscopy,” *Advanced Photonics Research*, vol. 6, no. 12, e202500180, **2025**. doi:10.1002/adpr.202500180
4. T. Kwon[†], **G. Song**[†], Y. Kim, J. Kim, J. C. Ye^{*}, M. Jang^{*}, “Video diffusion posterior sampling for seeing beyond dynamic scattering layers,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 12, 11152–11167, **2025**. doi:10.1109/TPAMI.2025.3598457

5. D. Lee[†], **G. Song**[†], C. Lee, C. Lee, M. Jang^{*}, “Reconstructive spectrometer using double-layer disordered metasurfaces,” *Science Advances*, vol. 11, no. 22, eadv2376, **2025**. doi:10.1126/sciadv.adv2376
6. J. You[†], D. Lee[†], **G. Song**, C. Lee, M. Jang, “Motion-free high-resolution on-chip microscopy using LED matrix,” *Optics Express*, vol. 32, no. 21, 36549–36562, **2024**. doi:10.1364/OE.537127
7. G.-H. Go, D. Lee, J. Oh, **G. Song**, D. Lee, M. Jang, “Meta Shack–Hartmann wavefront sensor with large sampling density and large angular field of view: phase imaging of complex objects,” *Light: Science & Applications*, vol. 13, no. 1, 187, **2024**. doi:10.1038/s41377-024-01528-9
8. A. Shibukawa, R. Higuchi, **G. Song**, H. Mikami, Y. Sudo, M. Jang, “Large-volume focus control at 10 MHz refresh rate via fast line-scanning amplitude-encoded scattering-assisted holography,” *Nature Communications*, vol. 15, no. 1, 2926, **2024**. doi:10.1038/s41467-024-47009-w
9. C. Lee, **G. Song**, H. Kim, J. C. Ye, M. Jang, “Deep learning based on parameterized physical forward model for adaptive holographic imaging with unpaired data,” *Nature Machine Intelligence*, vol. 5, no. 1, 35–45, **2023**. doi:10.1038/s42256-022-00584-3
10. J. Oh, C. Lee, **G. Song**, M. Jang, “Review of endomicroscopic imaging with coherent manipulation of light through an ultrathin probe,” *Journal of Optical Microsystems*, vol. 3, no. 1, 011004, **2023**. (Invited review) doi:10.1117/1.JOM.3.1.011004
11. H. Kim, **G. Song**, J. You, C. Lee, M. Jang, “Deep learning for lensless imaging,” *Journal of the Korean Physical Society*, vol. 81, no. 6, 570–579, **2022**. (Invited review) doi:10.1007/s40042-022-00412-2

Preprints

1. G.-H. Go, D. Lee, **G. Song**, M. Jang, “Random-mapped intensity optical neural network: all-optical two-layer computing for multimodal optical-field inference,” *arXiv*, 2609.02698, **2026**. doi:10.48550/arXiv.2609.02698

Conference Presentations

1. [Oral] **G. Song**, D. Lee, M. Jang, “Spectrometer using stacked disordered metasurfaces,” *Computational Optical Sensing and Imaging (COSI), Optica Imaging Congress · paper CTh1B.1*, Seattle, WA, USA, **Aug 2025**. — Student Paper Prize
2. J. Oh, H. Ko, **G. Song**, S. Lee, M. Jang, “Deep-tissue optical focusing with phase conjugation based on nonlinear acousto-optic modulation,” *CLEO/Pacific Rim · paper Th1G_2*, Incheon, Korea, **Aug 2024**.
3. [Poster] **G. Song**, D. Lee, C. Lee, C. Lee, M. Jang, “Double-layered disordered metasurface spectrometer,” *Korea International Semiconductor Conference on Manufacturing Technology (KISM 2023)*, Busan, Korea, **Nov 2023**.
4. [Oral] **G. Song**, M. Jang, “Resolution enhancement in ultrasound-modulated optical tomography by detecting high-order diffracted light waves,” *Optics and Photonics Congress (OPC 2021)*, Busan, Korea, **Jul 2021**.
5. [Oral] **G. Song**, D. Lee, M. Jang, “Second-harmonic detection of ultrasound-modulated light,” *32nd Annual Meeting of the Optical Society of Korea (OSK 2021 Winter)*, Korea, **Feb 2021**.

Patents & Inventions

- M. Jang, **G. Song**, C. Lee, D. G. Lee, “Ultrasensitive spectrometer,” US 12,571,712 B2 (granted Mar 10, 2026); KR 10-3018644 (granted Sep 8, 2026).
- M. Jang, **G. Song**, D. Lee, “Hyperspectral image sensor and system employing the same,” KR 10-2926493 (granted Feb 6, 2026); US 2024/0280406 A1 (pending Aug 22, 2024).
- **G. Song**, M. Jang, “Compact microscope based on a random-encoding metasurface,” KR 10-2025-0132777 (pending 2025).

Inventor since high school — Founded and led Sekwang High School's first invention and patenting club (2012–2014), ran workshops and guided members through filing. Three applications of my own: *Mechanism for contact-lens application* · KR 10-2013-0090667; *Repair tool for mechanical-pencil tips* · KR 20-2014-0009434; *Pen-cap holder cradle* · KR 20-2014-0009014.

Grants & Funded Projects

- Master's & Ph.D. Adventure Research · finalist, ₩10M** — KAIST College of Engineering 2021
- Student leader. A portable 3D fluorescence imager built on deep neural networks and scattering media, for miniaturised nucleic-acid diagnostics.
- Preliminary Startup Team Support · ₩9M** — KAIST Startup Support Foundation 2021
- Student leader. A portable nucleic-acid diagnostic device based on on-chip hyperspectral imaging.

Research Experience

- Beam shaping with metasurfaces · LG Electronics** 2025 – 2026
- Fabricated a wide-field-of-view metasurface for an ultra-compact projector.
 - Developed an efficient large-scale wave-propagation simulator.
- Computational imaging for semiconductor inspection · Samsung Electronics DS** 2022 – 2025
- Built a fully functional Fourier ptychographic microscope.
 - Developed aberration-correction algorithms and alignment protocols for industrial inspection.
 - Engineered inverse-design techniques for stacked-metasurface phase maps.
- Deep-tissue microscopy by spatiotemporal gating · National Research Foundation of Korea** 2021 – 2025
- Developed a differential-detection method for high-resolution ultrasound-modulated optical tomography.
 - Synchronised ultrasonic and optical pulses for spatiotemporal gating; 2 talks and an M.S. thesis.
- Snapshot hyperspectral sensing with 2.5D disordered metastructures · Samsung Future Technology Center** 2020 – 2025
- Designed a predictive sensor-response model for stacked disordered metasurfaces.
 - Built an ultrafast wavelength–space scanner with a supercontinuum laser for real-time hypercube acquisition.
 - Demonstrated snapshot hypercube measurement through scattering media; two patent families, now granted in the US and Korea.

Technical Skills

Cleanroom

- *Deposition*: PECVD (SiO₂, SiN_x), Thermal evaporation, E-beam evaporation
- *Lithography*: Photolithography, E-beam lithography, Soft lithography
- *Etching*: ICP-RIE, Wet etching, Lift-off

Optical bench

- *Holography*: Off-axis, Phase-stepping (EOM, AOM), Frequency-modulated (AOM + focused ultrasound), Time-gated (fs laser), Pulse-synchronised (ns laser + pulsed ultrasound)
- *Wavefront shaping*: Digital optical phase conjugation (DMD, SLM)
- *Systems built*: Fast-scanning monochromator, Hyperspectral object projector, Confocal microscope, Fourier ptychographic microscope

Code & simulation

- *Languages*: Python (PyTorch, TensorFlow), MATLAB, C/C++
- *Simulation*: FDTD, RCWA, Large-scale wave propagation

Prototyping

- *Modelling*: Fusion 360
- *Visualisation*: Blender, Maya, Illustrator
- *Fabrication*: FDM 3D printing, SLA/DLP 3D printing, Laser cutting